# The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform

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# 1

- : The Interplay of AI-and-RAN: Dynamic Resource Allocation for Converged 6G Platform
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- : IEEE INFOCOM 2025 Workshop (arXiv:2503.07420)
- : O-RAN, 5G, AI, xApp, RIC, SAC

### 1.1

AI-RAN Alliance **AI-and-RAN**(AI workload RAN workload ) , O-RAN NRT-RIC **CAORA** , GPU (MIG) workload **SAC(Soft Actor-Critic)** . RAN workload 99% RAN 100% .

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## 2 (Motivation)

RAN network function **dedicated** . (static) .

1. **Off-peak** : RAN .
2. **AI workload** : GenAI/edge AI inference .

workload .

| | RAN workload | AI workload |
|---|---|---|
| | latency-sensitive, (non-elastic) carrier ( ) | (elastic) use case best-effort |

, . **RAN** **AI workload** , **(coexistence)** .

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## 3 CAORA

**CAORA (Converged AI-and-ORAN Architecture)** O-RAN 4 .

### 3.1

- **Layer 1 — External Cloud**: AI/ML workload . edge AI inferencing, generative AI, robotics application , **AI Deployment API** RAN (compute cluster) .
- **Layer 2 — Programmable RAN**: O-RAN disaggregation O-RU / O-DU / O-CU.
- **Layer 3 — Controller (NRT-RIC)**: traffic prediction, resource allocation, monitoring custom xApp . **E2 interface** RAN .
- **Layer 4 — Resource Management (E2E Orchestrator)**: compute cluster(GPU, **MIG**, NIC) . SAC agent .

### 3.2 : Y1 interface

**Y1 interface** .

- **Monitoring xApp** (Layer 3) bandwidth utilization, system latency, network load RAN KPI .
- radio analytics **Y1 interface** E2E orchestrator(Layer 4) .
- E2E orchestrator **Y1 consumer** SAC .

*"xApp = , E2E orchestrator(SAC) = "* , SAC agent : (i) **(predictive analysis)**, (ii) **(real-time adaptation)**, (iii) peak RAN **(priority management)**.

RAN network AI workload **multi-purpose system** .

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# 4

## 4.1 Task

task $K$, task $k \in K$ :

$$k = \{\tau, r, p\}$$

- $\tau$: task (RAN / AI)
- $r = [r_i,\ \forall i = 1, 2, \ldots, N]$: (resource type )
- $p \in [0, 1]$: ( > / )

$r(k,t)$, $p(k)$ .

## 4.2

task $k$ $t$ $L_1$ norm .

$$\Delta(k,t) = \|r(k,t)\|_1 = \sum_{i=1}^{n} r_i(k,t)$$

task .

$$\Delta_{\text{total}}(t) = \sum_{k \in K} \Delta(k,t)$$

## 4.3 (System Dynamics)

workload $W(t)$ task , **(contention)** .

$$W(t) = \int_0^t \left( \sum_{k \in K} P(k,t)\,\Delta(k,t) \;-\; \sum_{k \in K} P_c(k,t)\,\Delta(k,t)\,C(k,t) \right) dt$$

- $P(k,t)$: task
- $P_c(k,t) = \alpha \cdot p(k)$: task ($\alpha$ , $p(k)$ )
- $C(k,t)$: task $k$

, ( ) .

## 4.4

RAN AI , . .

$$r_{\text{RAN}}(t) + r_{\text{AI}}(t) \;\le\; \mathcal{R}_{\max}$$

$\mathcal{R}_{\max}$ , NVIDIA MIG ( ).

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# 5 SAC

E2E orchestrator **SAC(Soft Actor-Critic)** . SAC entropy - , .

.

- **State**: monitoring xApp RAN/AI (bandwidth utilization, system latency, network load ).
- **Action**: RAN AI ( ).
- **Reward**: task , **(utilization)** . peak RAN .

*"RAN " "AI workload best-effort "* .

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# 6

## 6.1

- **Python** (O-RAN ).
- **NVIDIA MIG(Multi-Instance GPU)** . " " GPU instance .
- RAN/AI dynamic scenario .

## 6.2

- RAN workload **99%** — RAN .
- RAN **100%** — off-peak AI workload .
- Peak RAN SAC AI RAN → priority-weighted reward .
- AI workload RAN , .

, RAN .

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# 7

( ) , .

1. : RAN (L1/MAC) , MIG . dataset .
2. : task MIG → fractional/variable allocation (PRB/TB) .
3. **NRT-RIC** : Y1/E2 **10 ms~** . " " sub-TTI .
4. **AI workload** : AI task (token, attention ) " MIG " .

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# 8 Gap

AI-RAN **AI-and-RAN( / )** . **importance-aware MAC scheduling** **AI-for-RAN( )** , **·granularity** .

## 8.1

| (CAORA) | | |
|---|---|---|
| | NRT-RIC xApp + E2E orchestrator | GPU-collocated **L2 MAC scheduler** |
| | NRT-RIC (10 ms~ ), Y1/E2 | **sub-TTI**, |

| | (CAORA) | |
|---|---|---|
| | MIG (GPU ) | **PRB / TB** ( ) |
| AI workload | ( MIG ) | **token importance**(entropy · attention) scheduler utility |
| “AI-and-RAN” | GPU RAN+AI **compute** | offloaded VLM RAN **importance-aware** |
| | , | **end-to-end task utility per resource** |
| | Python | Sionna → cuMAC on GH200, ROS2 |

### 8.2

- **AI workload** : AI workload *GPU* . use case multi-robot VLM offloading *uplink* . “AI RAN GPU ” , “AI RAN ” .
- : NRT-RIC/Y1 , sub-TTI externalize (GPU-collocated MAC ) .
- : monitoring xApp , · **token** **proactive** **PRB** . .

### 8.3 Positioning

- predictive AI-and-RAN orchestration , **convergence backbone** , **macro** .
- (arXiv:2507.09124, *Proactive AI-and-RAN Workload Orchestration*) , reactive/ proactive gap anchoring .

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## 9

“ *RAN AI GPU(MIG)* ” NRT-RIC + SAC , AI-and-RAN .
: RAN ( VLM offloading) **token** **sub-TTI MAC** **importance-aware** — (micro) .